

Model with Rigid Prices

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Definition of rigid price:

- Price that moves in direction of flexible price (bargained price) but less than it

Rigid-price norm

$$p^n = \left[\frac{x^\varepsilon \cdot \mu}{k} \right]^\sigma \cdot p_0 \quad \text{with } \left\{ \begin{array}{l} \sigma \in [0, 1) \\ p_0 > 0 \end{array} \right.$$

AD parameters (pointing to x^ε)
AS parameter (pointing to k)

• $\sigma = 0$: price is fixed, $p^n = p_0$

• $\sigma = 1$: price is flexible

• if $p_0 = \frac{(1-\beta)^{\varepsilon-1}}{f(1-\beta)(\beta/(1-\beta))}$: price is surplus-sharing for β

Comparative statics:

Start from solution equation

$$y^d(x, p) = y^s(x)$$

$$\frac{x^\varepsilon}{[1+\tau(x)]^{\varepsilon-1}} \cdot \frac{\mu}{p} = y(x) \cdot k$$

$$\left[\frac{x^\varepsilon \cdot \mu}{k} \right] \cdot \frac{1}{p} = y(x) [1+\tau(x)]^{\varepsilon-1}$$

Insert price norm

$$\left[\frac{x^\varepsilon \mu}{k} \right] \cdot \left[\frac{x^\varepsilon \mu}{k} \right]^{-\sigma} \cdot \frac{1}{p_0} = \int(x) [1 + \tau(x)]^{\varepsilon-1}$$

$$\left[\frac{x^\varepsilon \mu}{k} \right]^{1-\sigma} \cdot \frac{1}{p_0} = \int(x) [1 + \tau(x)]^{\varepsilon-1}$$

defines π

Same definition of π as w/ fixed price p_0 except for exponent $1-\sigma > 0$

→ Comparative statics w/ rigid price are same as w/ fixed price

→ But effects are attenuated. elasticity of tightness wrt x, μ, k is $(1-\sigma) \times$ elasticity under fixed price.

